

KEN CALLAGHAN

Data Engineer | Automation & Infrastructure Specialist

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with 4+ years of experience architecting scalable data pipelines, automating mission-critical reporting workflows, and managing secure, high-availability database environments. Delivered measurable business impact including 50% reduction in report generation time and KPI frameworks that drive executive decision-making at a professional services firm. Deep expertise in Python (advanced), SQL optimization, Linux server administration (Arch Linux), Salesforce integration, and end-to-end infrastructure ownership. Proven ability to eliminate operational bottlenecks, ensure data integrity/security, and collaborate cross-functionally to deliver actionable insights. Ready to bring technical excellence and ownership mindset to a high-growth team paying market rate for proven impact.

TECHNICAL SKILLS

Languages:	Python (Advanced: Pandas, NumPy, Openpyxl, Flask, Requests, BeautifulSoup4, NLTK, Tkinter, PyQt, Hashlib, PyAutoGUI), R (Intermediate), SQL (Query Optimization, Stored Procedures), Bash, HTML/CSS
Data & DB:	Database Administration (Security, Backup/Recovery, Data Integrity, Query Tuning), ETL/ELT Pipelines, Automated Reporting Systems, KPI Framework Design, Data Visualization Backends, Salesforce Data Management
Infrastructure:	Linux Server Admin (Arch Linux), Self-Hosted Applications, Network Architecture (PoE, NETGEAR), Power Redundancy (APC UPS), Synology NAS, Technical SEO, Google Analytics 4, Git/GitHub, VS Code, Command Line Mastery
Tools:	Salesforce Developer Console, Microsoft Office Suite, Notion, Matplotlib, Seaborn, SQLite, PostgreSQL concepts

PROFESSIONAL EXPERIENCE

Data Engineer & Database Administrator

Jackson Cross Partners | Philadelphia, PA | March 2022 – Present

- Architected and deployed custom Python automation pipelines (Pandas + Openpyxl) that eliminated manual reporting bottlenecks, reducing report generation time by 50% and freeing 20+ hours/week for high-value analysis across the organization.
- Designed and implemented enterprise-grade KPI frameworks and scalable data visualization backends integrated with Salesforce, empowering leadership with real-time, actionable insights that directly influenced strategic business decisions and revenue growth initiatives.
- Optimized full-stack backend infrastructure and technical SEO, driving measurable improvements in site performance and search visibility tracked via Google Analytics 4 – contributing to increased organic lead generation and operational efficiency.
- Owned end-to-end database administration: implemented rigorous security protocols, automated backup/recovery strategies, data integrity monitoring, and SQL query optimization – ensuring 99.9%+ data availability, confidentiality, and compliance across all client projects.
- Engineered and maintained production self-hosted Linux server environments (Arch Linux) with custom networking (NETGEAR PoE), power redundancy (APC UPS), and application hosting – demonstrating deep infrastructure ownership and high-availability design principles.
- Partnered with cross-functional teams to pioneer innovative market research methodologies, translating complex datasets into clear recommendations that supported high-stakes business development and client strategy.

KEY PROJECTS

- **Home Server & Network Architecture (Production-Grade):** Designed and deployed a resilient self-hosted Synology NAS environment featuring automated power failover (APC UPS), enterprise-grade local networking with NETGEAR PoE switch, 4K security camera system, and custom Arch Linux configurations for monitoring,

diagnostics, and secure application hosting. Proves hands-on expertise in infrastructure-as-code thinking, redundancy planning, and full-stack systems ownership.

- **Enterprise Reporting Automation Framework:** Built reusable Python/Openpyxl pipeline that cut manual reporting effort by 50%. Framework adopted team-wide, directly improving productivity and data accuracy for ongoing operations.
- **P90X Community Platform:** Developed public-facing web repository using HTML/CSS integrated with Internet Archive to expand free access to fitness resources – demonstrating full-stack web development and passion for community-driven solutions.

EDUCATION

Kutztown University of Pennsylvania | 2014 – 2018

Relevant Coursework: Advanced Programming (Python, SQL, R), Twitter Bot Development, Mass Communication (TOR/FOSS/Internet Protocols), Data Analysis Foundations

WHAT I'M BRINGING TO YOUR TEAM

- Immediate productivity: I hit the ground running with production automation, DB optimization, and infrastructure skills that deliver ROI in weeks, not months.
- Proven cost-saver: My automation work has already saved organizations significant time/money – expect the same for you.
- Full ownership mindset: From code to servers to stakeholder communication, I take end-to-end responsibility.
- Market rate: After 4 years delivering at this level, I'm targeting \$115k–\$135k base + benefits in the Philly market (hybrid/remote friendly).